

HW3 - Generalized Linear Models

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1 Model Implementation

We implement multinomial and ordinal logistic regression in two ways and compare them.

1.1 Autograd + gradient descent

We built a scalar autograd engine (the `Value` class) supporting addition, multiplication, power, exp, and log, each with a `_backward` closure that implements the chain rule. Backpropagation walks a topological sort of the computation graph in reverse. Both models are trained by constructing the negative log-likelihood from `Value` operations and calling `backward()` to obtain gradients, then updating parameters with vanilla gradient descent.

1.2 Optimization library

We use `scipy.optimize.fmin_l_bfgs_b` with numerical gradients (`approx_grad=True`). The same negative log-likelihood is computed with NumPy arrays. No manual gradient derivation or autograd is needed.

1.3 Comparison

	Autograd + GD	L-BFGS-B (scipy)
Multinomial fit time	0.36 s	0.003 s
Ordinal fit time	0.24 s	0.003 s

Speed. L-BFGS-B is roughly $100\times$ faster. The scalar autograd has Python-level overhead per operation, whereas L-BFGS-B uses vectorized NumPy and a compiled optimizer with second-order curvature information.

Convergence. L-BFGS-B converges more reliably in fewer iterations thanks to approximate Hessian information. The autograd approach requires manual tuning of learning rate and epoch count; too large a step diverges, too small is slow.

Numerical stability. Both require the same softmax/sigmoid stability tricks (subtracting max, clipping probabilities). The autograd version is more fragile for ordinal regression, if thresholds are not initialized with proper spacing, intermediate probabilities become zero and `log` fails.

Prediction similarity. On the test data, predictions differ by at most 0.003 (multinomial) and 0.016 (ordinal), indicating both converge to similar optima.

Implementation difficulty. The autograd version requires substantially more code (building the `Value` class, manual training loop). The scipy version is compact, and of course already written for us.

2 Application

2.1 Multinomial Regression on Basketball Shots

We fit multinomial logistic regression on the basketball dataset (5024 shots, 6 shot types) using one-hot encoded categoricals and standardized continuous features. We estimated coefficient uncertainty via 20 bootstrap resamples; coefficients marked below are those where $|\hat{\beta}| > 2 \cdot \text{SE}$.

Key findings:

- **TwoLegged** is the strongest predictor. Two-legged shots strongly favor layups (+5.7) and dunks (+4.1) over hook shots (−3.6) and above-head (−3.9). This makes sense biomechanically: layups culminate in a two-footed gather, while hook shots are executed on one foot with a sweeping motion.
- **Movement** has large effects. Stationary play (no movement) massively increases hook shots (+8.7) and decreases layups (−10.1). Driving strongly favors layups (+3.9) and reduces above-head shots (−5.1). Players who drive to the basket finish close-range with layups; stationary post players use hooks.
- **Distance** separates shot types clearly. Farther shots favor above-head (+4.0) and other (+3.8), while close range favors tip-ins (−6.1) and dunks (−3.1). This is intuitive: you can only dunk or tip-in right at the rim.
- **Competition level:** U14 players almost never dunk (−5.2) or tip-in (−4.8), younger players lack the height and strength for these shots. They rely more on above-head shots (+2.6) and layups (+2.9).
- **Angle** has a modest effect: higher angles (more lateral) favor layups (+0.37) and disfavor tip-ins (−0.30). Shots from directly in front are less likely to be layups.
- **Transition** increases dunks (+0.67) and decreases hook shots (−0.66), fast breaks create easy close-range finishes rather than deliberate post moves.

Overall, the physical context (distance, stance, movement) dominates shot type selection, while competition level captures physical development differences.

2.2 Ordinal Regression: When It Outperforms Multinomial

Ordinal regression outperforms multinomial when the true data-generating process respects the proportional odds assumption: a single latent score shifts probability mass across ordered classes via shared thresholds.

Example DGP: Let $x \sim \mathcal{N}(0, 1)$ and $z = 2x + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 0.5)$. Assign classes via thresholds: $y = 0$ if $z < -1$, $y = 1$ if $-1 \leq z < 1$, $y = 2$ if $z \geq 1$.

Here the ordinal model matches the DGP exactly (one β , two thresholds), while multinomial wastes parameters fitting K separate weight vectors. With limited data (small n), multinomial overfits while ordinal generalizes better. The advantage grows with more classes K , since multinomial needs $O(pK)$ parameters vs. ordinal’s $O(p + K)$.

The function `multinomial_bad.ordinal_good` in `solution1.py` implements this generator.

3 GLM Diagnostics

We fit a simple linear regression predicting Distance from Angle and produced three diagnostic plots (Fig. 1). The model is extremely poor ($R^2 = 0.003$), and the diagnostics confirm this.

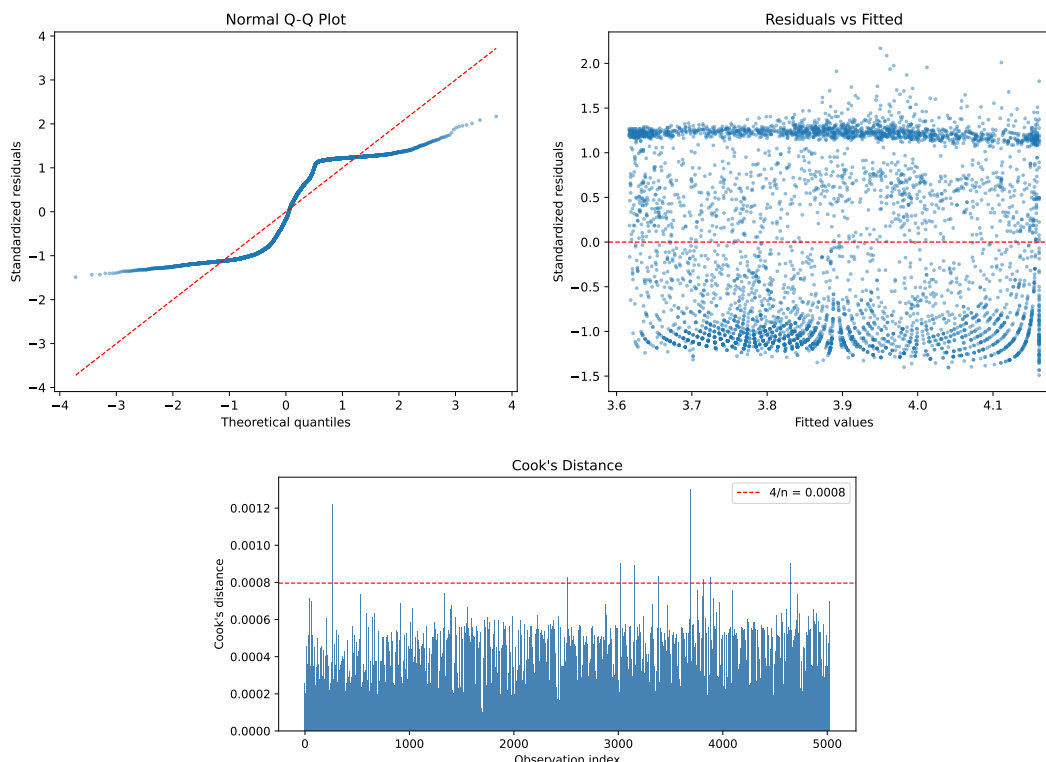


Figure 1: Diagnostic plots for $\text{Distance} \sim \text{Angle}$.

Q-Q plot: Heavy S-shape departure from the diagonal. The left tail flattens (residuals clipped around -1.2) and the right tail curves below the line. The residuals are clearly non-normal: the response (distance) is bounded below by 0 and right-skewed, which a normal linear model cannot capture.

Residuals vs fitted: Two distinct horizontal bands are visible rather than random scatter. The upper band near $+1.2$ and the lower spread indicate the response has a multimodal or discrete-like structure (distance values cluster at certain ranges depending on shot type). The banding pattern also shows heteroscedasticity: the spread is not constant across fitted values.

Cook's distance: No single observation dominates the fit; all Cook's distances are below 0.0013 , well under the threshold of 1 . Nine observations exceed $4/n = 0.0008$, but their influence is negligible. This is expected with $n = 5024$: individual points have little leverage in large samples.

Conclusion: The diagnostics indicate that the linear model is severely misspecified. The near-zero R^2 , non-normal residuals, and banded residual structure all suggest that the Angle-Distance relationship is not linear, and that Distance's distribution violates the normality assumption of ordinary linear regression.